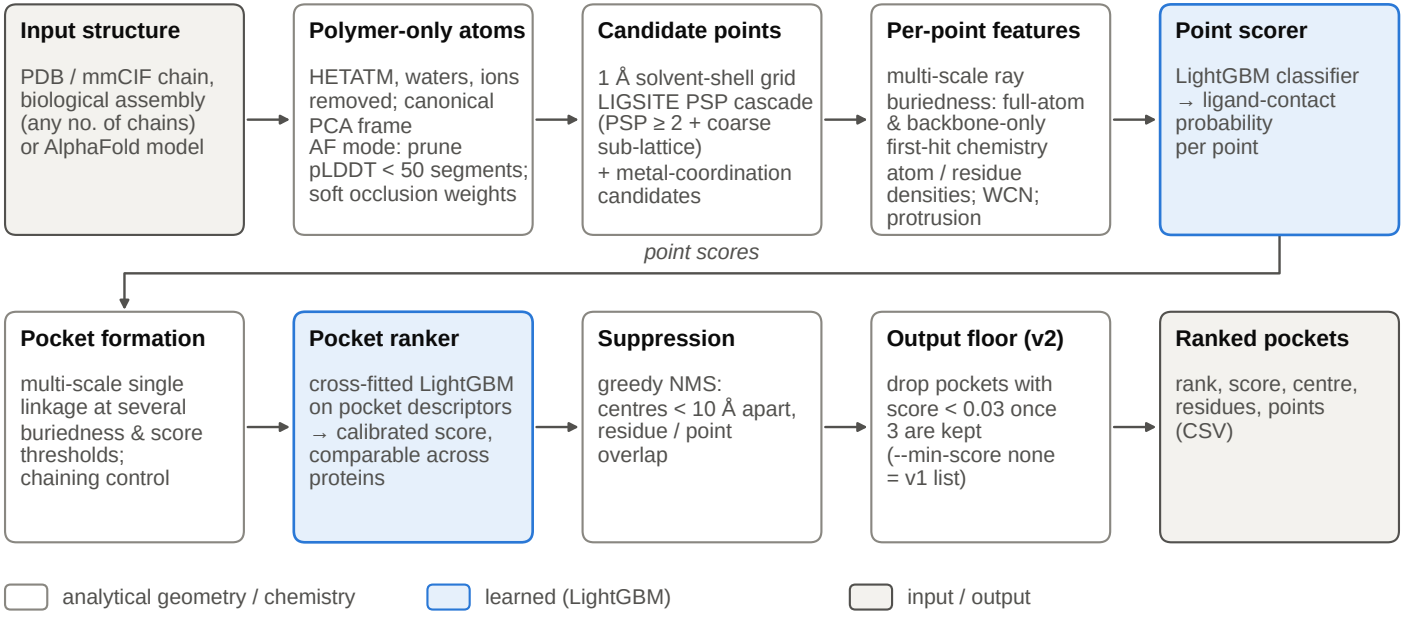
Input structure
PDB / mmCIF chain, biological assembly (any no. of chains) or AlphaFold model
Polymer-only atoms
HETATM, waters, ions removed; canonical PCA frame
AF mode: prune pLDDT < 50 segments; soft occlusion weights
Candidate points
1 Å solvent-shell grid
LIGSITE PSP cascade (PSP ≥ 2 + coarse sub-lattice)
+ metal-coordination candidates
Per-point features
multi-scale ray buriedness: full-atom & backbone-only
first-hit chemistry
atom / residue densities; WCN; protrusion
Point scorer
LightGBM classifier → ligand-contact probability per point
point scores
Pocket formation
multi-scale single linkage at several buriedness & score thresholds;
chaining control
Pocket ranker
cross-fitted LightGBM on pocket descriptors → calibrated score, comparable across proteins
Suppression
greedy NMS: centres < 10 Å apart, residue / point overlap
Output floor (v2)
drop pockets with score < 0.03 once 3 are kept
(--min-score none = v1 list)
Ranked pockets
rank, score, centre, residues, points (CSV)
analytical geometry / chemistry
learned (LightGBM)
input / output